Bird Watching

I observed the number of birds flying across my window in 30-minute intervals each day over 10 consecutive days. Each observation was made consistently from 5:00 to 6:30 PM, divided into three separate time windows: 5:00–5:30 PM, 5:30–6:00 PM, and 6:00–6:30 PM. By observing at the same times each day, I aimed to eliminate systematic variation due to time of day, ensuring the data is comparable across days and that any differences reflect randomness.

I chose this phenomenon because the movement of birds appeared naturally random, with no clear pattern or predictability. I saw several species of birds, and different species are likely to exhibit distinct behavioral patterns. The bird movements likely stem from various external factors, such as weather, noise levels, and the availability of food. Therefore, the randomness comes from environmental variability, individual behavior, and external disturbances.

The number of birds observed in the 30-minute intervals over the 10 days was as follows: {1, 2, 3, 5, 0, 7, 2, 1, 3, 2, 1, 7, 4, 0, 1, 4, 6, 6, 4, 2, 5, 3, 3, 2, 1, 6, 3, 2, 2, 3}

I hypothesize that the number of birds observed in each 30-minute window follows a **Poisson distribution** because:

Discrete Events: The number of birds crossing the window is a countable, discrete variable.
Fixed Interval: Observations were conducted consistently over fixed periods of 30 minutes each, ensuring a standard time frame for each count.
Independence: Each bird crossing the window is independent of others. This assumption is supported by the lack of noticeable flocking behavior and the presence of several different bird species. Their movements are likely influenced by their individual needs, such as searching for food, finding shelter, or having fun.
Constant Average Rate: Over the observation period, there appeared to be a roughly consistent average rate of bird crossings in each 30-minute interval, even though individual counts varied slightly.

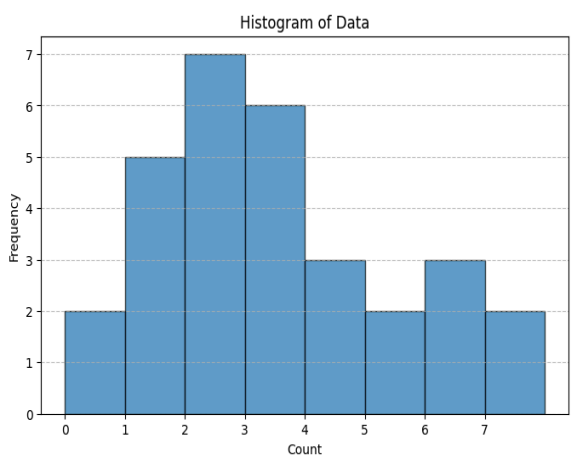


Figure 1: Histogram of the data. Data seems skewed to the right, which is in line with the Poisson distribution at $\lambda = 3.03$.

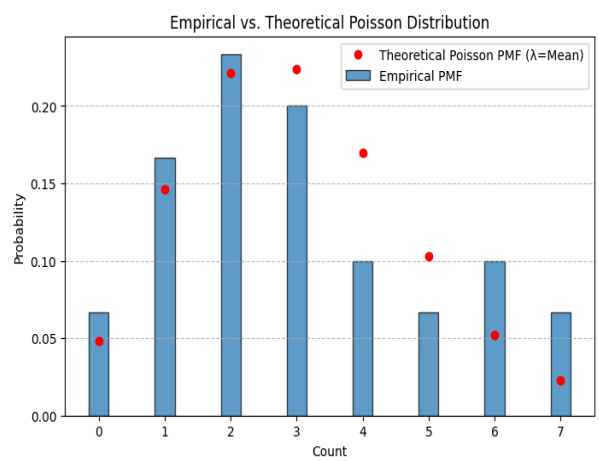


Figure 2: Empirical vs Theoretical Poisson Distribution. The shape of the probability distributions seems to match overall.

Mean (λ): $\frac{Sum\ of\ observations}{Number\ of\ days} = \frac{91}{30} \approx 3.03$ **Variance(σ^2)**: $\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n} \approx 3.83$

$P_{empirical}(k) = \frac{Count\ of\ occurrences\ of\ k}{Total\ number\ of observations}$

The close alignment between the mean and variance supports the hypothesis that the data follows a Poisson distribution. In Figure 2, we see that the shape of the probability distributions seems to match overall between the empirical and theoretical.

Bayesian Inference: Let's consider the Poisson, Normal, and Uniform (discrete) distributions to determine which one best describes the observed data. I'm considering the Normal distribution because the data might approximate a bell curve if bird crossings cluster around a central value with variability but note that it is a continuous distribution. I'm considering the Uniform distribution because it assumes all counts within a range are equally likely, which could serve as a baseline model in the absence of clear patterns or trends in the data.

$$P(M_i|data) = \frac{P(data|M_i)\cdot P(M_i)}{P(data)} = \frac{P(data|M_i)\cdot P(M_i)}{\Sigma_i P(data|M_i)\cdot P(M_i)}$$

Due to my previous analysis, my initial belief is that the data likely follows the Poisson distribution. Therefore, the priors are $P(M_{Poisson}) = \frac{2}{3}$, $P(M_{Normal}) = \frac{1}{6}$, $P(M_{Uniform}) = \frac{1}{6}$.

Likelihoods: Since the data points are assumed to be independent, the total likelihood is the product of individual probabilities for each data point.

$P(data|M_{Poisson}) = \prod_{x \in data} P(X = x|\lambda) = \prod_{x \in data} \frac{\lambda^x e^{-\lambda}}{x!}$ where x is the observed count of birds in the 30-minute intervals and λ is the mean of the data.

$P(data|M_{Normal}) = \prod_{x \in data} P(X = x|\mu, \sigma) = \prod_{x \in data} \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$ where μ is the mean and σ is the standard deviation of the data.

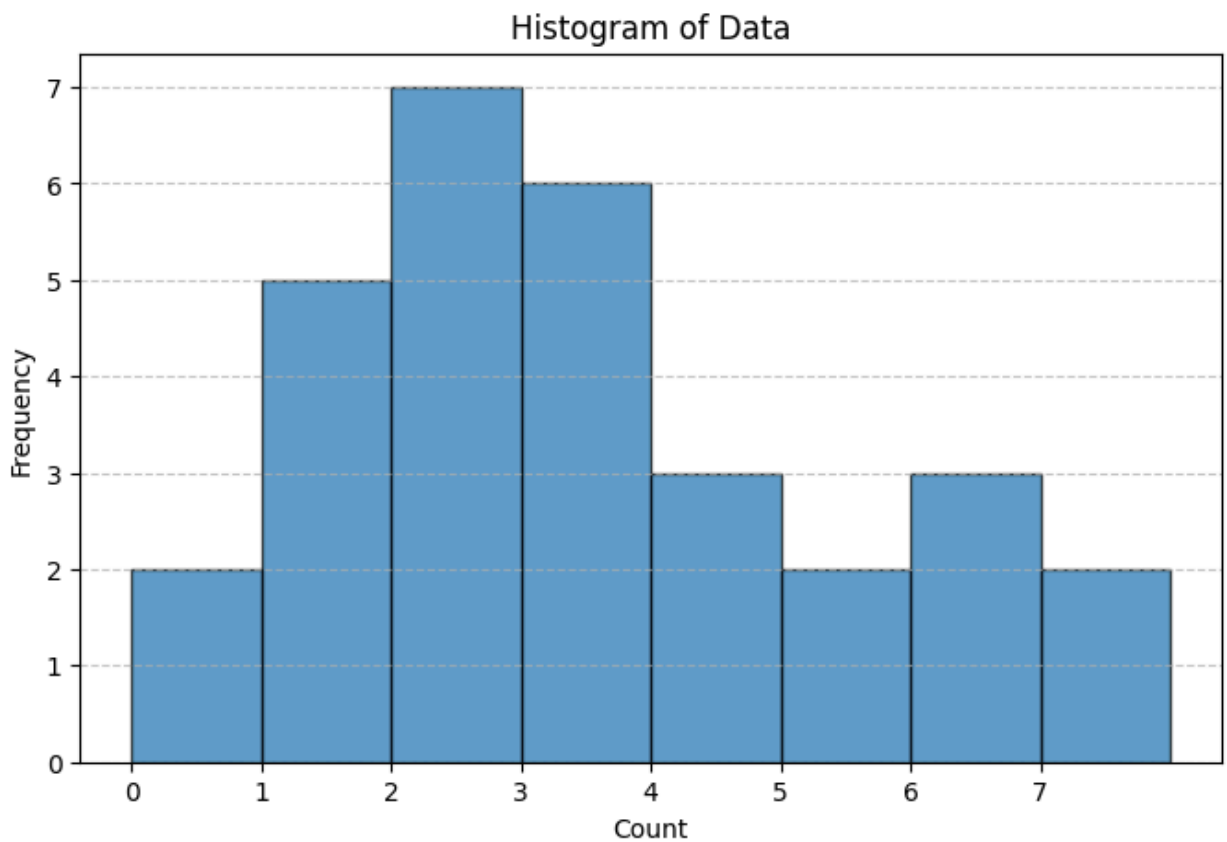
$P(data|M_{Uniform}) = \prod_{x \in data} P(X = x|a, b) = (\frac{1}{b-a+1})^n$ where a and b are the minimum and maximum values in the data and n is the number of data points. After calculating, we find the posteriors are 0.15, 0.01, and 0.84, respectively.

The results suggest that the Uniform distribution is the most probable model given the observed data. This result highlights the flexibility of the Uniform distribution, especially with a small sample size, as it does not penalize variability or deviations as heavily as structured models like Poisson or Normal. It assumes no clustering or patterns, which the small data size favors. Despite this, the close alignment between the empirical data and the Poisson model reinforces its validity as a plausible fit, suggesting that with a larger dataset, the Poisson distribution may provide stronger evidence.

Appendix

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
data = [1, 2, 3, 5, 0, 7, 2, 1, 3, 2, 1, 7, 4, 0, 1, 4, 6, 6, 4, 2, 5, 3, 3, 2, 1, 6,
3, 2, 2, 3]
mean_data = np.mean(data)
variance_data = np.var(data)
print(f"Mean: {mean_data}, Variance: {variance_data}")

plt.figure(figsize=(8, 5))
plt.hist(data, bins=range(min(data), max(data) + 2), edgecolor='black', alpha=0.7)
plt.title("Histogram of Data")
plt.xlabel("Count")
plt.ylabel("Frequency")
plt.xticks(range(min(data), max(data) + 1))
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

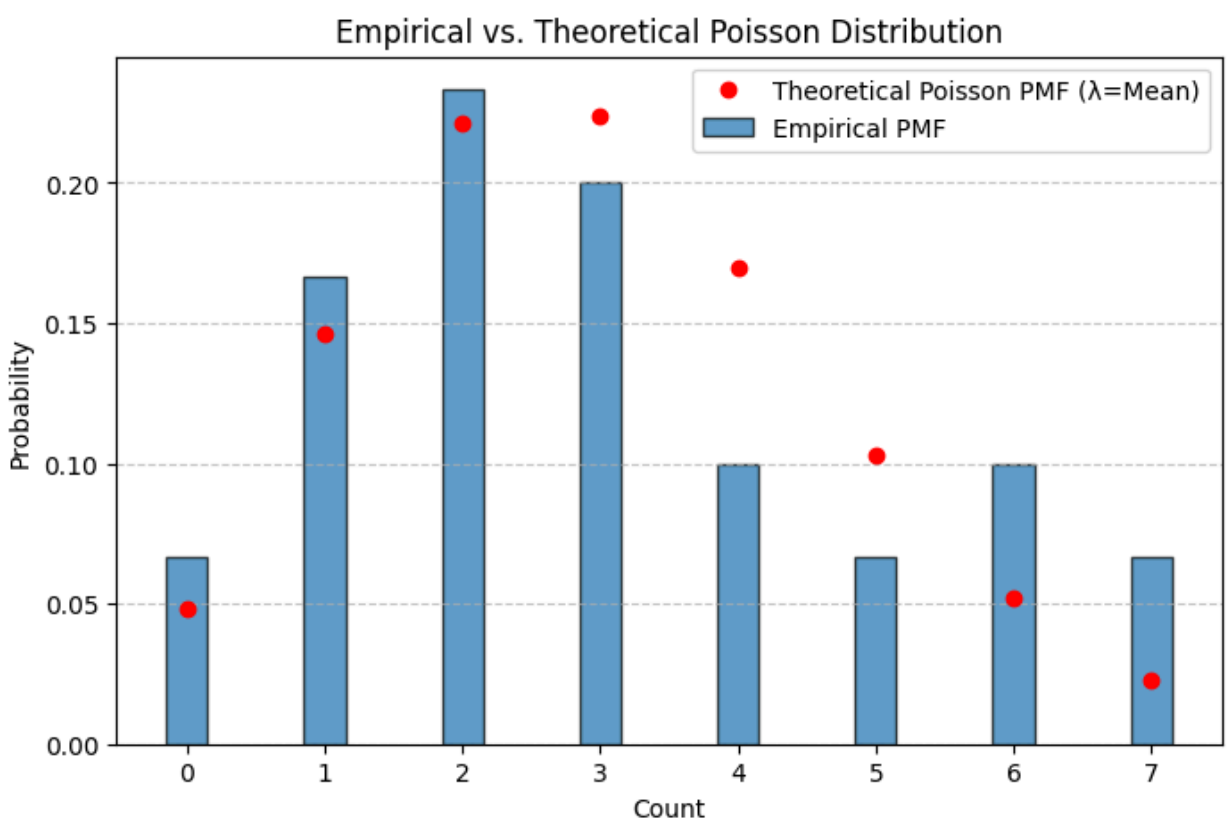



```
from scipy.stats import poisson
```

```
# Unique counts in the data and empirical probabilities
unique_counts = range(min(data), max(data) + 1)
empirical_pmf = [data.count(x) / len(data) for x in unique_counts]

# Theoretical Poisson PMF
poisson_pmf = [poisson.pmf(x, mean_data) for x in unique_counts]

plt.figure(figsize=(8, 5))
plt.bar(unique_counts, empirical_pmf, width=0.3, label="Empirical PMF",
edgecolor='black', alpha=0.7)
plt.plot(unique_counts, poisson_pmf, 'o', color="red", label="Theoretical Poisson PMF
(λ=Mean)")
plt.title("Empirical vs. Theoretical Poisson Distribution")
plt.xlabel("Count")
plt.ylabel("Probability")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.legend()
plt.show()
```




```
from scipy.stats import poisson, norm
```

```python
import numpy as np

data = [1, 2, 3, 5, 0, 7, 2, 1, 3, 2, 1, 7, 4, 0, 1, 4, 6, 6, 4, 2, 5, 3, 3, 2, 1, 6,
3, 2, 2, 3]
mean_data = np.mean(data)
variance_data = np.var(data)

prior_poisson = 2/3
prior_normal = 1/6
prior_uniform = 1/6

likelihood_poisson = np.prod([poisson.pmf(x, mean_data) for x in data])
likelihood_normal = np.prod([norm.pdf(x, mean_data, np.sqrt(variance_data)) for x in
data])

min_data = min(data)
max_data = max(data)
range_size = max_data - min_data + 1
likelihood_uniform = (1 / range_size) ** len(data)  # All values equally likely within
the range

evidence = (likelihood_poisson * prior_poisson +
           likelihood_normal * prior_normal +
           likelihood_uniform * prior_uniform)

posterior_poisson = (likelihood_poisson * prior_poisson) / evidence
posterior_normal = (likelihood_normal * prior_normal) / evidence
posterior_uniform = (likelihood_uniform * prior_uniform) / evidence

print(f"Posterior probability that data follows Poisson distribution:
{posterior_poisson}")
print(f"Posterior probability that data follows Normal distribution:
{posterior_normal}")
print(f"Posterior probability that data follows Uniform distribution:
{posterior_uniform}")
```

```
Posterior probability that data follows Poisson distribution: 0.8468408233094183
Posterior probability that data follows Normal distribution: 0.06381221934818895
Posterior probability that data follows Uniform distribution: 0.08934695734239262
```